

Topical Therapy for Rosacea

Several topical agents have demonstrated efficacy in clearing inflammatory papules and pustules and reducing erythema when applied once daily. Twice-daily application or combination therapy may be required if monotherapy is insufficient.

- **Metronidazole** and **azelaic acid** are classified as pregnancy category B.
- **Sodium sulfacetamide/sulfur** is pregnancy category C.
- Azelaic acid may cause initial tingling or burning, which typically resolves with continued use.
- Sodium sulfacetamide/sulfur **medicated cleansers** are better tolerated in patients with sensitive skin compared to "leave-on" formulations, which may exacerbate stinging and burning.
- Daily use of a **barrier repair emollient** is recommended to support skin integrity.

Off-Label Topical Treatments

Common off-label topical agents include:

- **Benzoyl peroxide**: Effective for papules and pustules but may irritate sensitive skin.
- **Clindamycin**: Twice-daily topical clindamycin has shown superior pustule clearance compared to oral tetracycline in one study.
- **Erythromycin**: Used topically, though evidence is limited.
- **Calcineurin inhibitors** (tacrolimus ointment, pimecrolimus cream): Most effective for steroid-induced perioral dermatitis but may benefit select rosacea patients, particularly those with inflammatory lesions.

Adjunctive Manual Therapy

Facial massage, performed according to **Soybe's technique**, may serve as an adjunctive treatment:

- Begins at the central face (glabella and nose).
- Involves sweeping motions toward the inferolateral face (mandibles and lateral neck) in the direction of lymphatic flow.
- Aims to mobilize edema and accelerate clearance of dermal inflammation.

Topical Retinoids

- **Tretinoin cream** promotes connective tissue remodeling and reduces dermal inflammation over time.
- Small clinical studies support the use of topical retinoids in rosacea.
- Clinical improvement is delayed, typically requiring **4 to 6 months** of consistent use.
- Potential for irritation and theoretical concerns about angiogenesis have limited their use, though long-term data do not support an increased risk of telangiectasia.
- Retinoids inhibit **vascular endothelial growth factor (VEGF)** production via anti-AP1 activity in keratinocytes.
- Tolerability can be improved by:
 - Gradual introduction of the retinoid.
 - Concomitant use of barrier repair emollients.
 - Particularly valuable for **long-term maintenance** therapy.

Oral Therapy for Rosacea

Topical management is generally preferred to minimize risks of antimicrobial resistance and systemic side effects. However, oral therapies are used off-label and can provide rapid symptom control, especially in moderate to severe cases.

General Considerations

- Oral antibiotics should be used short-term; long-term maintenance should rely on topical agents.
- Photosensitizing agents require caution due to the photodistributed nature of rosacea.
- All discussed oral therapies are used **off-label**.

Tetracyclines

- Effective for reducing papules, pustules, and erythema.
- Have been used since the 1950s at both antimicrobial and sub-antimicrobial doses.
- Faster onset of action compared to isotretinoin.
- Relapse rates:
 - ~25% within 1 month of discontinuation.
 - 50% by 6 months off therapy.
- **Topical maintenance** is recommended after oral treatment.
- **Contraindicated** in pregnant women and those planning pregnancy.

Isotretinoin

- Low-dose isotretinoin (10–40 mg daily, or <0.5 mg/kg/day) is effective and better tolerated in rosacea patients.
- One small study showed:
 - Significant reduction in facial blood flow (measured by laser-Doppler) with 30 mg daily for 10 weeks.
 - No significant change with tetracycline 250 mg twice daily over the same period.
- Useful for short-term control of moderate to severe flushing or erythema.